

# Model Question Paper-I with effect from 2025

USN

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1BMATC101

## First Semester B.E./B.Tech. Degree Examination Differential Calculus and Linear Algebra

TIME: 03Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**  
 2. VTU Formula Hand Book is Permitted  
 3. M: Marks, L: Bloom's level, C: Course outcomes

| Module-1        |   |                                                                                                                                                                                                                                                                                                  | M | L  | CO |
|-----------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|
| Q 1.            | a | With usual notations, prove that $\tan(\phi) = r \frac{d\theta}{dr}$ .                                                                                                                                                                                                                           | 6 | L2 | 1  |
|                 | b | Find the angle between pair of curves, $r \sec^2\left(\frac{\theta}{2}\right) = 2a$ ,<br>$r \operatorname{cosec}^2\left(\frac{\theta}{2}\right) = 2b$ .                                                                                                                                          | 7 | L2 | 1  |
|                 | c | Find the pedal equation of the curve $r^n = a^n \cos n\theta$ .                                                                                                                                                                                                                                  | 7 | L2 | 1  |
| <b>OR</b>       |   |                                                                                                                                                                                                                                                                                                  |   |    |    |
| Q 2.            | a | Find the radius of curvature of the curve $x^3 + y^3 = 3axy$ , at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ .                                                                                                                                                                          | 6 | L2 | 1  |
|                 | b | Find the radius of curvature of the parametric curve $x = a(t - \sin t)$ , $y = a(1 - \cos t)$ .                                                                                                                                                                                                 | 7 | L2 | 1  |
|                 | c | Show that $\rho$ varies inversely as $r^{n-1}$ for the curve $r^n = a^n \sin(n\theta)$ ,                                                                                                                                                                                                         | 7 | L2 | 1  |
| <b>Module-2</b> |   |                                                                                                                                                                                                                                                                                                  |   |    |    |
| Q 3.            | a | Obtain the series of the function $y = \log(1 + \cos x)$ upto the term contains $x^4$                                                                                                                                                                                                            | 6 | L2 | 1  |
|                 | b | Evaluate (i). $\lim_{x \rightarrow a} \left( \frac{x^x - a^x}{x^a - a^a} \right)$ and (ii). $\lim_{x \rightarrow 0} \frac{\sinh(x) - x}{\sin(x) - x \cos(x)}$ .                                                                                                                                  | 7 | L2 | 1  |
|                 | c | If $u = x^3 - 3xy^2 + x + e^x \cos y + 1$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$                                                                                                                                                                  | 7 | L2 | 1  |
| <b>OR</b>       |   |                                                                                                                                                                                                                                                                                                  |   |    |    |
| Q 4.            | a | If $z = f(x, y)$ where $x = e^u \sin v$ and $y = e^u \cos v$ then prove that $\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = e^{2u} \left\{ \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right\}$ | 6 | L2 | 1  |
|                 | b | If $u = \frac{yz}{x}$ , $v = \frac{xz}{y}$ , $w = \frac{xy}{z}$ show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$                                                                                                                                                                      | 7 | L2 | 1  |
|                 | c | Find the extreme value of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$                                                                                                                                                                                                                     | 7 | L2 | 1  |
| <b>Module-3</b> |   |                                                                                                                                                                                                                                                                                                  |   |    |    |
| Q 5.            | a | Solve the differential equation $\cos x \, dy = y(\sin x - y)dx$                                                                                                                                                                                                                                 | 6 | L2 | 1  |

|                 |          |                                                                                                                                                                                                                                                                                               |          |           |          |
|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|----------|
|                 | <b>b</b> | Solve the differential equation $(x - y)dx - dy = 0$ , $y(0) = 2$                                                                                                                                                                                                                             | <b>7</b> | <b>L2</b> | <b>1</b> |
|                 | <b>c</b> | Solve the differential equation<br>$(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$                                                                                                                                                                                                                | <b>7</b> | <b>L2</b> | <b>1</b> |
| <b>OR</b>       |          |                                                                                                                                                                                                                                                                                               |          |           |          |
| <b>Q 6.</b>     | <b>a</b> | Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$ where $\lambda$ is the parameter.                                                                                                                                                                       | <b>6</b> | <b>L2</b> | <b>1</b> |
|                 | <b>b</b> | A bacterial population B is known to have a rate growth proportional to B itself. If between noon and 2PM the population triples, at what time no controls being exerted, should B become 100times what it was at noon.                                                                       | <b>7</b> | <b>L3</b> | <b>1</b> |
|                 | <b>c</b> | Radium decomposes at a rate proportional the quantity of radium present. Suppose that it is found that in 25 years approximately 1.1% of a certain quantity of radium has decomposed. Determine approximately how long will it take for half of the original amount of radium to decompose. - | <b>7</b> | <b>L3</b> | <b>1</b> |
| <b>Module-4</b> |          |                                                                                                                                                                                                                                                                                               |          |           |          |
| <b>Q 7.</b>     | <b>a</b> | Solve $y^{111} - 3y^{11} - y^1 + 3y = 0$                                                                                                                                                                                                                                                      | <b>6</b> | <b>L2</b> | <b>1</b> |
|                 | <b>b</b> | Solve $(D^2 - D + 1)y = \sinh x$                                                                                                                                                                                                                                                              | <b>7</b> | <b>L2</b> | <b>1</b> |
|                 | <b>c</b> | Solve $(D^2 + 5D - 6)y = \sin 4x \sin x$                                                                                                                                                                                                                                                      | <b>7</b> | <b>L2</b> | <b>1</b> |
| <b>OR</b>       |          |                                                                                                                                                                                                                                                                                               |          |           |          |
| <b>Q 8.</b>     | <b>a</b> | Solve $(D^2 + 1)y = \cos x$ by the Method of variation of parameters.                                                                                                                                                                                                                         | <b>6</b> | <b>L2</b> | <b>1</b> |
|                 | <b>b</b> | Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} 5y = x^2 \cos x (\log x)$                                                                                                                                                                                                                     | <b>7</b> | <b>L2</b> | <b>1</b> |
|                 | <b>c</b> | Construct the governing equation of the free, damped motion of a mass?                                                                                                                                                                                                                        | <b>7</b> | <b>L2</b> | <b>1</b> |
| <b>Module-5</b> |          |                                                                                                                                                                                                                                                                                               |          |           |          |
| <b>Q 9.</b>     | <b>a</b> | Find the rank of the matrix $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & 7 \end{bmatrix}$                                                                                                                                                        | <b>6</b> | <b>L2</b> | <b>2</b> |
|                 | <b>b</b> | Test for the consistency of the following system and then solve $x-2y+3z=2$ , $3x-y+4z=4$ , $2x+y-2z=5$ .                                                                                                                                                                                     | <b>7</b> | <b>L2</b> | <b>2</b> |
|                 | <b>c</b> | Solve the following system of equation by Gauss elimination method<br>$2x_1 + 2x_2 + x_3 + 2x_4 = 7$ , $-x_1 + 2x_2 + x_4 = -2$ ,<br>$-3x_1 + x_2 + 2x_3 + x_4 = -3$ , $-x_1 + 2x_4 = 0$ .                                                                                                    | <b>7</b> | <b>L3</b> | <b>2</b> |
| <b>OR</b>       |          |                                                                                                                                                                                                                                                                                               |          |           |          |
| <b>Q 10.</b>    | <b>a</b> | Solve the following system of equations by Gauss seidel method<br>$10x + y + z = 12$ , $x + 10y + z = 12$ , $x + y + 10z = 12$                                                                                                                                                                | <b>6</b> | <b>L3</b> | <b>2</b> |
|                 | <b>b</b> | Apply power method to find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & 6 \\ 3 & 6 & -2 \end{bmatrix}$ .                                                                                                                                             | <b>7</b> | <b>L3</b> | <b>2</b> |
|                 | <b>c</b> | Diagonalize the matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ .                                                                                                                                                                                                                   | <b>7</b> | <b>L2</b> | <b>2</b> |

# CBCS SCHEME

USN

1 V 3 2 5 C V 0 0 2

1BMATC101

## First Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Differential Calculus and Linear Algebra : CV Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks, L: Bloom's level, C: Course outcomes.  
3. VTU Handbook is permitted.*

| Module - 1 |    |                                                                                                                                                                          | M | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | With usual notations, prove that $\tan \phi = r \frac{d\theta}{dr}$ .                                                                                                    | 6 | L2 | CO1 |
|            | b. | Find the angle of intersection of the curves $r^2 \sin 2\theta = 4$ and $r^2 = 16 \sin 2\theta$                                                                          | 7 | L2 | CO1 |
|            | c. | Show that for the curve $r(1 - \cos \theta) = 2a$ , $\rho^2$ varies as $r^3$ .                                                                                           | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                                          |   |    |     |
| Q.2        | a. | With usual notations, prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left( \frac{dr}{d\theta} \right)^2$ .                                                   | 6 | L2 | CO1 |
|            | b. | Find the pedal equation of the polar curve $r^m = a^m (\cos m\theta + \sin m\theta)$                                                                                     | 7 | L2 | CO1 |
|            | c. | Show that the radius of curvature at (a,0) on the curve $y^2 = \frac{4a^2(2a-x)}{x}$ is 'a'.                                                                             | 7 | L2 | CO1 |
| Module - 2 |    |                                                                                                                                                                          |   |    |     |
| Q.3        | a. | Expand $\log(\sec x + \tan x)$ in powers of x as far as the term in $x^5$ .                                                                                              | 6 | L2 | CO1 |
|            | b. | If $u = f\left\{\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right\}$ , then show that $\sum x \frac{\partial u}{\partial x} = 0$                                               | 7 | L2 | CO1 |
|            | c. | If $u = \frac{yz}{x}, v = \frac{xz}{y}, w = \frac{xy}{z}$ , show that $\frac{(u,v,w)}{(x,y,z)} = 4$ .                                                                    | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                                          |   |    |     |
| Q.4        | a. | Evaluate i) $\lim_{x \rightarrow 0} \left[ \frac{a^x + b^x + c^x}{3} \right]^{\frac{1}{x}}$ ii) $\lim_{x \rightarrow 0} \left[ \frac{\sin x}{x} \right]^{\frac{1}{x^2}}$ | 6 | L2 | CO1 |
|            | b. | If $u = f(y-z, z-x, x-y)$ then show that $\sum \frac{\partial u}{\partial x} = 0$                                                                                        | 7 | L2 | CO1 |
|            | c. | Find the extreme value of the function $f = x^4 + y^4 - 2(x-y)^2$                                                                                                        | 7 | L2 | CO1 |
| Module - 3 |    |                                                                                                                                                                          |   |    |     |
| Q.5        | a. | Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$                                                                                          | 6 | L2 | CO1 |
|            | b. | $[4xy + 3y^2 - x]dx + x(x + 2y)dy = 0$                                                                                                                                   | 7 | L2 | CO1 |
|            | c. | A bacterial culture growing exponentially increases from 100 to 400 grams in 10 hours. How much was present after 1 hour                                                 | 7 | L3 | CO1 |

| OR         |    |                                                                                                                                                                                                                                                                                                      |   |    |     |  |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|--|
| Q.6        | a. | Solve $xy(1+xy^2)\frac{dy}{dx} = 1$                                                                                                                                                                                                                                                                  | 6 | L2 | CO1 |  |
|            | b. | Find the Orthogonal trajectories of the family of curve $r = a(1 + \sin\theta)$                                                                                                                                                                                                                      | 7 | L2 | CO1 |  |
|            | c. | In a certain chemical reaction the rate of conversion of a substance at time 't' is proportional to the quantity of the substance still untransformed at that instant. At end of 1 hour 60grams remains and at the end of 4 hours 21grams. how many grams of the first substance was there initially | 7 | L3 | CO1 |  |
| Module - 4 |    |                                                                                                                                                                                                                                                                                                      |   |    |     |  |
| Q.7        | a. | Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = e^{2x} + \sin x + x$                                                                                                                                                                                                                               | 6 | L2 | CO1 |  |
|            | b. | Solve by variation of parameter method $\frac{d^2y}{dx^2} + 4y = \tan 2x$                                                                                                                                                                                                                            | 7 | L2 | CO1 |  |
|            | c. | Solve $(x+1)^2 \frac{d^2y}{dx^2} + (x+1) \frac{dy}{dx} + y = 2\sin\{\log(1+x)\}$                                                                                                                                                                                                                     | 7 | L2 | CO1 |  |
| OR         |    |                                                                                                                                                                                                                                                                                                      |   |    |     |  |
| Q.8        | a. | Solve $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$                                                                                                                                                                                                                             | 6 | L2 | CO1 |  |
|            | b. | Solve $x^2 \frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + 6y = x$                                                                                                                                                                                                                                            | 7 | L2 | CO1 |  |
|            | c. | Solve by variation of parameter method $\frac{d^2y}{dx^2} + y = \sec x$                                                                                                                                                                                                                              | 7 | L2 | CO1 |  |
| Module - 5 |    |                                                                                                                                                                                                                                                                                                      |   |    |     |  |
| Q.9        | a. | Find the rank of the matrix $\begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 1 \end{bmatrix}$                                                                                                                                                                         | 6 | L2 | CO2 |  |
|            | b. | Solve the following system of equation by Gauss elimination method<br>$x - 2y + 3z = 2$ , $3x - y + 4z = 4$ , $2x + y - 2z = 10$ .                                                                                                                                                                   | 7 | L3 | CO2 |  |
|            | c. | Find the Eigen values and Eigen vectors of the following matrix by using Rayleigh's power method. $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$                                                                                                                         | 7 | L2 | CO2 |  |
| OR         |    |                                                                                                                                                                                                                                                                                                      |   |    |     |  |
| Q.10       | a. | Investigate the values of $\lambda$ and $\mu$ such that<br>$2x + y + 4z = 12$ , $x + 2y + 3z = 10$ , $x + 2y + \lambda z = \mu$ .<br>(i) Unique Solution (ii) Infinite Solution (iii) No Solution                                                                                                    | 6 | L3 | CO2 |  |
|            | b. | Solve by Gauss-seidel iteration method:<br>$83x + 11y - 4z = 9$ , $3x + 8y + 29z = 71$ , $7x + 52y + 13z = 104$                                                                                                                                                                                      | 7 | L3 | CO2 |  |
|            | c. | Find the eigen values and eigen vectors of the matrix<br>$A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$                                                                                                                                                                                     | 7 | L2 | CO2 |  |

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**First Semester B.E/B.Tech. Degree Examination, June/July 2026**  
**Differential Calculus and Linear Algebra: CV Stream**

Time: 3 hrs.

Max. Marks:100

**Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
 2. M : Marks , L: Bloom's level , C: Course outcomes.  
 3. VTU Hand Book is permitted.

| Module - 1 |    |                                                                                                                                   | M | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 1          | a. | Prove with usual notation $\tan \theta = r \frac{d\theta}{dr}$                                                                    | 6 | L3 | CO1 |
|            | b. | Obtain the angle between the curves $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$ .                                       | 7 | L2 | CO1 |
|            | c. | Find the radius of curvature for the Folium of De-Cartes $x^3 + y^3 = 3axy$ at the point $(3a/2, 3a/2)$ on it.                    | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                   |   |    |     |
| 2          | a. | Show that the following pairs of curves intersect each other orthogonally $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$ . | 6 | L3 | CO1 |
|            | b. | Find the pedal equation of the curve $\frac{2a}{r} = (1 + \cos \theta)$                                                           | 7 | L2 | CO1 |
|            | c. | Show that for the curve $r = a(1 + \cos \theta)$ $\frac{\rho^2}{r}$ is constant.                                                  | 7 | L2 | CO1 |
| Module - 2 |    |                                                                                                                                   |   |    |     |
| 3          | a. | Using Maclaurin's series prove that $\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}$ .               | 6 | L3 | CO1 |
|            | b. | Evaluate $\lim_{x \rightarrow 0} \left[ \frac{a^x + b^x + c^x + d^x}{4} \right]^{1/x}$                                            | 7 | L2 | CO1 |
|            | c. | Show that the function $f(x, y) = x^3 + y^3 - 3xy + 1$ is minimum at $(1, 1)$ .                                                   | 7 | L3 | CO1 |

| OR         |    |                                                                                                                                                                                                                         |   |        |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
| 4          | a. | Expand $e^{\sin x}$ up to the term containing $x^4$ using Maclaurin's series.                                                                                                                                           | 6 | L3     |
|            | b. | If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ , prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$ .                                   | 7 | L2 CO1 |
|            | c. | If $u = \frac{yz}{x}$ , $v = \frac{xz}{y}$ , $w = \frac{xy}{z}$ , then show that $J\left(\frac{u, v, w}{x, y, z}\right) = 4$                                                                                            | 7 | L3 CO1 |
| Module - 3 |    |                                                                                                                                                                                                                         |   |        |
| 5          | a. | Solve : $x \frac{dy}{dx} + y = x^3 y^6$                                                                                                                                                                                 | 6 | L2 CO1 |
|            | b. | Solve : $(5x^4 + 3x^2 y^2 - 2xy^3) dx + (2x^3 y - 3x^2 y^2 - 5y^4) dy = 0$ .                                                                                                                                            | 7 | L3 CO1 |
|            | c. | Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$ , where $\lambda$ is a parameter.                                                                            | 7 | L3 CO1 |
| OR         |    |                                                                                                                                                                                                                         |   |        |
| 6          | a. | Solve: $(x^2 + y^2 + x)dx + xy dy = 0$ .                                                                                                                                                                                | 6 | L2 CO1 |
|            | b. | Find the orthogonal trajectories of $y^2 = 4ax$ , where $a$ is parameter .                                                                                                                                              | 7 | L3 CO1 |
|            | c. | A bacterial population B is known to have a rate growth proportional to B itself. If between noon and 2PM the population triples, at what time no controls being exerted, should B become 100times what it was at noon. | 7 | L3 CO1 |
| Module - 4 |    |                                                                                                                                                                                                                         |   |        |
| 7          | a. | Solve $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$ .                                                                                                                                                                             | 6 | L2 CO1 |
|            | b. | Solve $\frac{d^2 y}{dx^2} + y = \sec x \tan x$ using the method of variation of parameters.                                                                                                                             | 7 | L3 CO1 |

c. Solve  $x^3 \frac{d^3y}{dx^3} + 3x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$

7

L3

CO1

OR

8 a. Solve  $x^2 y'' + 4xy' + 2y = x^2$

6

L3

CO1

b. Solve by the method of variation of parameters  $\frac{d^2y}{dx^2} + a^2y = \sec ax$

7

L3

CO1

c. Solve the Legendre's linear equation  
 $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = 2 \sin \log(1+x)$

7

L3

CO1

## Module – 5

9 a. Find the rank of the matrix  $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$   
by elementary row transformations

6

L2

CO2

b. Solve the following system of equations by Gauss-Elimination method.  
 $x + y + z = 9$ ;  $x - 2y + 3z = 8$ ;  $2x + y - z = 3$ .

7

L3

CO2

c. Find the largest eigenvalue and the corresponding eigenvector

or of the matrix  $A$  by the power method, given that  $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$

7

L2

CO2

OR

10 a. Determine the values of  $\lambda$  and  $\mu$  such that the system of equations  
 $x + y + z = 6$ ;  $x + 2y + 2z = 10$ ;  $x + 2y + \lambda z = \mu$  may have a) unique solution, b) infinite number of solutions, c) no solution.

6

L2

CO2

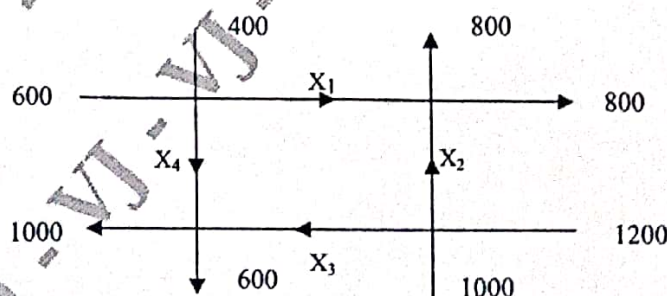
b. Solve the following system of equations by Gauss-Seidal method  
 $20x + y - 2z = 17$ ,  $3x + 20y - z = -18$ ,  $2x - 3y + 20z = 25$

7

L3

CO2

c. Write the system of linear equations of the traffic flow in the net of one way street directions as shown in the figure and find its solution.



7

L3

CO2

\*\* 3 of 3 \*\*